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Innovative Applications of O.R.

Stochastic modeling of integrated order fulfillment processes with delivery time promise: Order picking, batching, and last-mile delivery

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ABSTRACT

To guarantee high customer service and short and accurate lead times, many e-commerce retailers have started to home deliver their customer orders within a few hours or even minutes, also known as quick-commerce order fulfillment. Quick-commerce order fulfillment consists of three main processes: order picking in the warehouse, order batching for delivery, and last-mile delivery. The ultimate delivery performance depends on managing all three processes, which are highly stochastic, and interdependent. We capture this stochasticity and interdependency in an integrated analytical framework and derive approximate analytical expressions for the mean and variance of the total order fulfillment time. We validate the analytical expressions with both in-house detailed process simulations and external-party output measures. We then analyze the delivery cost-service quality trade-offs using an optimization model that minimizes the expected order fulfillment cost with a *delivery probability* (DP) constraint, focusing on meeting delivery time deadlines. The optimization model determines the number of pickers, the optimal delivery batch size, and the number of vehicles required to deliver the customer orders. Achieving a high delivery reliability comes at a cost. In comparison to the model with DP constraints, we observe that the expected order fulfillment cost averaged over all data parameter settings obtained from the model without DP constraints is 8.9% lower; however, the mean and standard deviation of order fulfillment time increase by 44.1% and 18.6%, respectively, which results in low delivery reliability. We further demonstrate that an integrated analysis of the order fulfillment process is essential to set reliable fulfillment due times.

1. Introduction

Global retail e-commerce sales amounted to US$ 5 trillion in 2021 and are projected to exceed US$ 7 trillion in 2025 (Coppola, 2023). Online retail growth has picked up even greater momentum during COVID-19 which has led to a surge in online retail, and customers now expect even faster delivery of their orders. E-commerce retailers have accepted these challenges by reducing order fulfillment time so that items are immediately picked, packed, and delivered to customers within the promised due time also known as "quick-commerce" order fulfillment or flash delivery.

Quick-commerce (q-commerce) focuses on delivering products quickly and efficiently to customers; typically, within hours or minutes from the order placement (Gund & Daniel, 2023; Rai et al., 2023; Waßmuth et al., 2023). Q-commerce has become popular during the pandemic, especially with grocery and essential products delivery services. In 2021, a Statista survey revealed that 47.6% of online shoppers in the US use rapid grocery delivery services. Further, customers demand both low-cost and fast service. While 60.6% respondents demand low (or no) delivery fees and 53.4% of respondents demand low item prices, 40.4% of respondents use this service for its fast delivery feature (Beyrouthy, 2023). Average order fulfillment times for q-commerce service vary from 10 min to 60 min; for example, companies such as Dija (UK), Gorillas (Germany), Getir (Turkey), and Zepto (India) offer delivery within 10 min, companies like Buyk (US) and Cajoo (France) offer a 15-minute delivery, companies such as Dunzo (India) offer a 19 min service, Gopuff (US) offer a 30 min service (Chevalier, 2022), and Flipkart Quick offers a 30–60 min service (Murali, 2020). To make this happen, these companies fully control all parts of the delivery process.

Typically, q-commerce retailers have a dedicated last-mile fulfillment center (micro-warehouse also known as a dark store that typically range between 250 to 5000 square feet) to fulfill the orders of a

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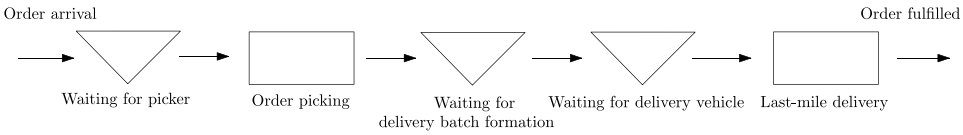


Fig. 1. Order fulfillment process in q-commerce.

particular service region. The order fulfillment time is driven by three major sub-processes: order picking in the warehouse, order batching for delivery, and last-mile delivery. By guaranteeing a short fulfillment lead time or by providing customers with accurate fulfillment time estimates, q-commerce companies drive customer satisfaction. A recent survey revealed that 14% respondents will abandon shopping with a retailer altogether if they receive a late delivery just one time (Hughes, 2023). Today express deliveries are also picking up at e-commerce giants such as Amazon who are expanding their same-day delivery network with items shipped from nearby fulfillment centers (Herrington, 2023). Same-day facilities with small footprints are designed for speedy operations and the fulfillment times are significantly faster than those of traditional fulfillment centers. Herrington (2023) remarks that the average time from picking up a customer's items to positioning the customer's package on the outbound dock is 11 min in Same-day facilities. BBExpress in India also rolled out 60-minute express deliveries.

For both q-commerce and same-day express delivery companies, offering speedy delivery and adhering to delivery time promises is crucial. Though speedy deliveries increase the warehousing and delivery manpower costs, a typical customer is unwilling to spend more on online purchases in comparison to offline purchases or pay a faster delivery premium. As a result, the provider's margins are quite low. Hence, the costs need to be controlled while breaches of delivery promises need to be monitored. Companies use analytical and predictive models to make reliable estimates of warehouse manpower and delivery personnel quantities, and order fulfillment times. However, capturing the uncertainty in demand and order processing times and providing reliable estimates of fulfillment time variance typically is challenging. By providing appropriate fulfillment time targets, retailers can deliver on their promises to their customers. Analytical models that can capture the variability in the order fulfillment processes can help us to understand the effect of handling and delivery capacities on order fulfillment times.

Fig. 1 depicts the flow of a customer order through various stages of the order fulfillment process. First, the customer order waits to be assigned to a picker in the storage area. The picker then retrieves all the line items in the order from the storage area and delivers them to the dispatch staging area. The orders are then batched and assigned to a delivery vehicle. Once the delivery vehicle is available, the batch of orders leaves the fulfillment center for last-mile delivery.

The order fulfillment time depends on several factors. These include order characteristics, such as the number of order lines per order as well as design characteristics, such as the number and length of aisles and cross aisles that directly affect the order pick times. In addition, operational design parameters, such as the number of order pickers, the order batch size (which may depend on delivery vehicle capacity), and the number of vehicles used for delivery have an impact. There are also interactions between the order picking, batching, and delivery processes, and customers expect both speed and low delivery costs. Therefore, we pose the following research question: How can a retailer adhere to the order delivery time targets and attain quality standards at acceptable costs in q-commerce? Increasing the number of resources such as order pickers or delivery vehicles can speed up the delivery, but also increases fixed operating costs such as the cost of order picking and delivery vehicles. How can a provider manage these cost-quality trade-offs in q-commerce?

To analyze these questions, we develop a two-stage methodology. In the first stage, we develop an integrated analytical model of the q-commerce order fulfillment process that comprises three phases: order picking, batching, and delivery. This model helps to estimate the mean and variance of the order fulfillment time as a function of the design parameters. In the second stage, we develop an optimization model to understand the cost-quality trade-offs in order fulfillment. While we model both waiting and resource costs for all three processes, we express the service quality of the order fulfillment process using the delivery probability (DP), which is the probability of fulfilling the customer order before the due time promised by the q-commerce or same-day express delivery company. Note that variability reduction in delivery lead times is crucial for integrated business processes to achieve high delivery service quality. Garg et al. (2006) develop an approach that facilitates variability reduction and business process synchronization for supply chains in a cost-effective way. They use delivery sharpness DS which, when used in combination with the classical performance index DP, measures the reliability and precision with which products are delivered to customers. Since q-commerce (also express delivery) companies promise a target delivery time to the customer, we choose DP as our service quality metric that denotes the proportion of orders delivered within a target time.

In this paper, we first develop a stochastic model for the integrated order fulfillment process, consisting of order picking, order batching, and order delivery, to derive analytical expressions for both the mean and variance of each process. We calculate the order pick time for any number of warehouse blocks, and the last-mile delivery time for any number of delivery zones. We integrate the three fulfillment processes using an open queuing network. From this network, we determine the waiting time of an order for the picker and for a delivery vehicle. We use the resulting performance estimates in an optimization model that minimizes the order fulfillment cost, with the number of pickers, the delivery batch size, and the number of delivery vehicles as decision variables. We further validate the model results with the external output data from a q-commerce grocery delivery company.

The model can handle trade-offs between the decision variables. For example, as customer orders are delivered in batches, the order waiting time to form the delivery order batch depends on the batch size. The smaller the batch size, the shorter the waiting time to form the delivery batch. However, a smaller batch size increases the number of delivery trips required to deliver all customer orders. Thus, there is a trade-off between the delivery batch size and the number of delivery trips. Furthermore, the order waiting time for the delivery vehicle depends on the number of delivery vehicles in service i.e., a smaller number of delivery vehicles leads to longer order waiting times in the delivery queue. However, a higher number of delivery vehicles leads to higher operational costs. In summary, we contribute to the literature and practice as follows:

1. We contribute to theory by developing an integrated analytical model to analyze multi-line order picking, order batching for delivery, and order delivery processes in q-commerce, able to handle stochasticity in the arrivals and the processes and taking into account delivery reliability. We are the first to develop such a model. As part of the model, we develop a travel time expression for the variance of picking multi-line orders in a multi-block warehouse setting, which is further used to determine the variance of order delivery time.

2. We contribute to e-retail practice by demonstrating the value of an integrated analytical model in setting order fulfillment time targets. To do so, we develop a decision support framework that helps set operational parameters, such as the number of pickers, delivery batch size, and the number of delivery vehicles needed to achieve a due time target. It also helps to understand the cost-service quality trade-off in the presence of a DP constraint.

The rest of the paper is organized as follows. Section 2 reviews the related literature on order picking system and last-mile delivery. Section 3 provides an open queuing network of the integrated fulfillment center processes and Section 4 analyzes the order picking, batching for delivery, and the last-mile delivery processes. Section 5 provides an optimization model to determine the optimal order batch size for delivery and the number of delivery vehicles while minimizing the expected order fulfillment cost with a DP constraint. In Section 6, we present an application of our model to a hyperlocal grocery delivery service company and validate the model against data obtained from an external party (a quick service grocery delivery company based out of India) and managerial insights from our work. In Section 7, we discuss the effect of order fulfillment resources and DP on the optimal decisions. Finally, conclusions and directions for future research are presented in Section 8.

2. Literature review

Our work analyzes the performance of integrated order-picking and last-mile delivery processes. There are two streams of literature in this area: (1) solving integrated order picking (order batching and pick sequencing) and vehicle routing problems using optimization formulations, and (2) analyzing performance and estimating throughput times of integrated order fulfillment processes using descriptive queuing and machine learning tools. We review these two areas in this section.

2.1. Solving integrated order picking and delivery planning problem

Zhang et al. (2016) study the integrated order picking and delivery problem for e-tailers for a storage area with a single block and multiple aisles. They formulate an integer program to maximize the number of orders delivered in the shortest possible time and consider a simple distribution model instead of a vehicle routing formulation for the delivery of customer orders. Moons et al. (2018) and Mackert (2019) provide an integer program for the integrated order picking and delivery problem using a vehicle routing formulation for the delivery of customer orders. They conclude that integrating order picking and delivery leads to a 14% reduction in cost. Kuhn et al. (2021) study a similar problem and propose an adaptive large neighborhood search algorithm to solve the integrated order picking and delivery problem. Ostermeier et al. (2021) study the integrated zone picking and vehicle routing problem with restricted buffer space. Rijal et al. (2021) examine a generic warehouse layout and minimize total work time by scheduling the order pickers, taking into account realistic constraints on the time needed for breaks and picker availability. Subsequently, Rijal et al. (2023) examine an integrated order fulfillment problem and combine order batching, picking, staging, and order delivery processes. They develop an optimization model to minimize the order fulfillment time and propose a dynamic programming algorithm to solve it. Schubert et al. (2021) develop a decision support tool to solve the integrated order batching, picker routing, picker scheduling, and deliveries with site dependencies (i.e., certain vehicle types can serve a particular customer class). They solve large instances using a variable neighborhood descent algorithm. Chen et al. (2022) solve the integrated scheduling of zone picking and vehicle routing problems with time windows using a Two-stage Iterated Search algorithm. They attempt to minimize operations and delay penalty costs. Jamili et al. (2022) develop an Integer Linear Programming (ILP) formulation towards integrated planning of the order picking and shipping process with an objective to minimize the total tardiness of shipping trucks. They also address the benefits of collaboration among order pick and dock door resources. The above-mentioned studies typically use all process times as deterministic inputs to the optimization model and obtain an integrated plan to attain a given objective such as minimize delay costs, or delivery tardiness. While our research also investigates the integrated order picking and delivery processes, we develop a stochastic model to estimate the throughput time for the integrated order fulfillment process.

2.2. Analyzing the performance of order picking and delivery processes

Order picking is the most labor-intensive process and the most costly function in order fulfillment (De Koster et al., 2007). We consider a picker-to-parts order pick system, where the picker travels to the pick locations to complete the order (De Koster, 2004). Studies on estimating the overall order fulfillment time in such a system, using stochastic models, are quite limited. There are studies on estimating isolated order picker travel times with different warehouse block layouts and another stream on estimating the total order picking throughput time.

Order pick time estimation: Many researchers have estimated the time the picker takes to retrieve the line item(s) from the storage area. Bassan et al. (1980) estimate the mean travel time to pick a line item in a route, when line item locations are uniformly distributed in the storage area. They also determine the optimal length-to-width ratio of the storage area for single-line orders. Hall (1993) estimates the mean travel time to pick multiple line items in a single route. He assumes that the line items are uniformly distributed in a storage area with multiple aisles and one block. Hall (1993) and Thomas and Meller (2014) analyze the effect of the number of line items per order on the optimal length-to-width ratio. They consider a single block storage area with a fixed number of line items in all the orders. Chew and Tang (1999) study a storage area with multiple aisles and one block and estimate the mean and variance of the order picking time, where the picker picks multiple line items in a single route. Later, Le-Duc and De Koster (2007) consider a storage area with multiple aisles and two blocks, and the picker picks multiple line items in a route. They extend the idea of Chew and Tang (1999) to estimate the mean and variance of order picking time. Roodbergen et al. (2008) consider a storage area with multiple aisles and blocks and estimate the mean of the order picking time. In their study, the picker can pick multiple line items in a route. We need to analyze the integrated process to estimate both the mean and variance of the order fulfillment time and the resulting delivery capability. Further, in q-commerce, especially for grocery delivery, orders can have multiple line items. The literature on estimating the variance of order picking time for multi-line orders is limited to a single or two block warehouse (Chew & Tang, 1999; Le-Duc & De Koster, 2007). We study storage areas consisting of multiple blocks and have to develop expressions estimating the travel time variance for such multi-block storage areas.

Order throughput time estimation: (Yu & De Koster, 2009) study order batching and pick area zoning. They develop approximations for the throughput time using two two-moment approximations of the order batching, picking, and sorting processes. Gong and De Koster (2008) provide an estimate for the mean of order waiting time in a dynamic order-picking system. Yu and De Koster (2008) model the order picking system with stochastic order picking time, where the picker picks only one line item in a trip. They use Whitt's queuing network analyzer and develop the first and second moments of order picking time. Recently, Tappia et al. (2019) and Bansal and Roy (2021) considered integrated analysis of item tote storage and retrieval (as the upstream process) and order picking as the downstream process. However, these studies focus only on the order-picking process (and there are many more that do so). However, these studies do not link the output of the order-picking process with the order deliveries. Further, their models neglect nonlinear waiting effects due to capacity utilization and process

variability in different order fulfillment processes. Another stream of literature leverages machine learning algorithms as a predictive approach for delivery time estimation. For example, Salari et al. (2022) develop a data-driven framework to predict the distribution of order delivery time and set promised delivery time to customers. Their rules are tested on JD.com data and they show sales improvement of 6.1% with their proposed delivery time promise policy. Likewise, Kandula et al. (2021) propose a decision support framework that can improve delivery success rates while reducing delivery costs. Using machine learning models, they generate delivery schedules by predicting the appropriate delivery time periods for order delivery. Our work differs, as we use a descriptive queuing-theoretic approach to due time estimation.

3. Modeling the order fulfillment process

The order fulfillment process covers all processes from when a customer places an order to delivery. To fulfill the orders, the fulfillment center needs to pick, batch, and deliver them. An order arrives at the picking station and waits until a picker becomes available (we focus on q-commerce delivery where pickers typically pick a single order). Once a picker is assigned to a particular order, the picker picks the required items from the shelf and transfers the completed order to the batching station where the orders are batched for delivery. The batch of orders waits for a vehicle to become available for last-mile delivery. The order fulfillment time can be divided into five components: (1) order waiting time for the picker, (2) order picking time, (3) order batching time for delivery, (4) batch waiting time for the delivery vehicle, and (5) delivery time. In the next Section 3.1 the system assumptions are introduced. Section 3.2 discussed the overall modeling approach. The details of deriving the means and standard deviations of all processing times are discussed in Section 4. Table 1 provides a list of notations used in this paper.

3.1. System assumptions

We consider an integrated system that consists of a fulfillment center delivering orders to a service region. The fulfillment center has a rectangular block layout, i.e. parallel aisles intersected by multiple cross aisles, to shortcut travel distances, and a depot at the end of the first aisle (see Fig. 3 and Appendix A). To analyze the order fulfillment performance, we make the following assumptions: (1) The line items are uniformly distributed over the locations in the storage area. This is a common assumption in the literature (see, e.g., Chew and Tang (1999), Le-Duc and De Koster (2007)) to estimate order picking time. This is in line with many e-commerce storage facilities. However, it is possible to assume different storage policies, albeit at the expense of more complicated travel time expressions. (2) The speed of the pickers is constant (Chew & Tang, 1999; Le-Duc & De Koster, 2007) while picking the items required for the order from the storage area. (3) The picker picks exactly one order at a time and retrieves all the line items corresponding to the order in a single trip in the storage area. This is a common approach in practice, in particular for online grocery item picking, where the orders are fairly large. A picker picks multiple line items in a trip. We use a data set from Instacart (a grocery delivery and pick-up service based in the US, The Instacart Online Grocery Shopping Dataset (2017)). In this data set, an order contains an average of ten line items. (4) A picker can move in both directions within aisles and cross-aisles and pickers can overtake each other. (5) The arrival of picked orders at the batching station follows a renewal process. (6) The customer locations are uniformly distributed in the rectangular service region with a grid structure (length l' and width l). This is a common approach in the literature (Daganzo, 1984a, 1984b) to estimate order delivery time. (7) The speed of the delivery vehicle is constant. (8) To determine the variance of order fulfillment time, we assume that the order picking and delivery processes are independent of each other. Typically, picking and delivery processes are performed by separate people and sometimes even separate service providers. Some of the above assumptions can be relaxed, e.g., the assumption of uniform pick locations and customer locations. However, the resulting travel time expressions become less transparent. As this does not add additional insights, we leave this for possible future research.

3.2. Integrated queuing model

The order fulfillment process can be modeled by an open queuing network as shown in Fig. 2. The customer orders arrive at the fulfillment center with rate λ and the coefficient of variation of the inter-arrival time is c_a. The order picking process is modeled as a $GI/G/N$ queue, where N is the number of pickers. The orders arrive at the order picking station and wait in the order picking queue until they are assigned to a picker. The mean and coefficient of variation of order picking time are μ_p^{-1} and c_p, respectively. The expected waiting time in the picking queue is given by Eq. (1) (Sakasegawa, 1977),

$$W_r = \left[\frac{U_p^{\sqrt{2(N+1)}-1}}{(1-U_p)N}\,\frac{c_a^2+c_p^2}{2}\right]\mu_p^{-1}, \tag{1}$$

where $c_p = \frac{\sigma_p}{\mu_p^{-1}}$ and $U_p = \frac{\lambda\mu_p^{-1}}{N}$.

The coefficient of variation of waiting time (c_r) in the order picking queue is as follows (Zhao et al., 2014):

$$c_r = \sqrt{1 + \frac{4\left(1 - P(T_r > 0)(\psi_r + 2)\right)}{3P(T_r > 0)(\psi_r + 1)}}, \tag{2}$$

where $\psi_r = \frac{1}{c_p^2}$, T_r is the random variable denoting the time an order waits in the order picking queue, and $P(T_r > 0)$ is the probability that an order waits in the order picking queue. The expression for $P(T_r > 0)$ is as follows (Zhao et al., 2014):

$$P(T_r > 0) = \frac{r^{\psi_r} c_p(1-c_a) + (\lambda/\mu_r')(1-c_p)c_a}{1 - c_p c_a}, \tag{3}$$

where $\mu_r' = \frac{\lambda}{(N\mu_p - \lambda)W_{rM/M/N}}$. $W_{rM/M/N}$ is the waiting time in an $M/M/N$ order picking queue and r is determined by the real root (< 1) of the equation (Zhao et al., 2014),

$$\mu_r' r^{\psi_r+1} - (\psi_r\lambda + \mu_r')r + \psi_r\lambda = 0$$

Once an order is assigned to a picker, the picker retrieves all the line items corresponding to the order and delivers them to the batching station. The mean and variance of the time taken by the picker to retrieve the items and to deliver them to the batching station are represented as μ_p^{-1} and σ_p^2, respectively. The estimation of the mean and variance of the order picking time is discussed in Section 4. The order departure rate and the coefficient of variation (CV) of order inter-departure time from the picking station are represented as λ and c_{pb}, respectively. Using Eq. (4), we calculate c_{pb} as follows (Whitt, 1983):

$$c_{pb}^2 = 1 + (1 - U_p^2)(c_a^2 - 1) + \frac{U_p^2}{\sqrt{N}}(c_p^2 - 1). \tag{4}$$

The order waits at the batching station until the batch is complete. After batch formation, the batch of orders moves to the order delivery station. The order batching for delivery is modeled as a queue. At the batching station, the order arrival rate is λ and the coefficient of variation of order inter-arrival time is c_{pb}. The mean (μ_b^{-1}) and variance (σ_b^2) of order batching time for delivery are estimated in Section 4. The departure rate from the batching station is λ/K and the coefficient of variation of inter-departure time from the batching station is $c_{pb}/\sqrt{K}$, where K is the order batch size for delivery. The batch of orders waits in the delivery station queue until it is assigned to a delivery vehicle. The mean of the time to deliver a batch of orders is μ_o^{-1} and CV of time deliver the order is $c_d/\sqrt{K}$, which are described in more detail in Section 4. The order delivery process is modeled as a $GI/G/S$ queue,

Table 1
Notations used in the model.

Process	Notation	Description
General	(t, c_a)	Mean and coefficient of variation of order inter-arrival time
	λ	Order arrival rate
	DP	Delivery probability
	E_{ft}	Expected order fulfillment time
	V_{ft}	Variance of order fulfillment time
	C	Expected order fulfillment cost
	D_u	Target for due time set by the company
	d	Target for DP set by the company
Order picking process	p_m	Probability of an order with m line items
	M_0	Maximum number of line items in an order
	N	Number of pickers in the storage area
	u	Speed of picker
	(μ_s^{-1}, σ_s^2)	Mean and variance of the picking time of a line item taken by the picker in the storage area
	$(\mu_p^{-1}, \sigma_p^2, c_p)$	Mean, variance, and coefficient of variation of order picking time which includes retrieval of all line items from the storage area and delivery of the order to the workstation
	(W_r, σ_r^2, c_r)	Mean, variance, and coefficient of variation of waiting time of the order in order picking queue
	U_p	Utilization of pickers in the storage area
	y_p	Picking cost per unit time per picker
Order batching process for delivery	K	Order batch size for delivery
	$(\mu_b^{-1}, \sigma_b^2, c_b)$	Mean, variance, and coefficient of variation of order batching time for delivery
	(σ_{pb}^2, c_{pb})	Variance and coefficient of variation of inter-arrival time of orders at the batching station
Order delivery process	A	Area of the service region for delivery
	$(\mu_d^{-1}, \sigma_d^2, c_d)$	Mean, variance, and coefficient of variation of order delivery time
	(μ_o^{-1}, σ_o^2)	Mean and variance of delivery time of an order batch
	(μ_t^{-1}, σ_t^2)	Mean and variance of the time taken by the customer to collect the order from the delivery personnel
	(μ_l^{-1}, σ_l^2)	Mean and variance of the time required to load a batch of orders on the delivery vehicle
	δ_d	Distance traveled by delivery vehicle to deliver a batch of orders
	(W_q, σ_q^2, c_q)	Mean, variance, and coefficient of variation of waiting time of the order in the order delivery queue
	S	Number of delivery vehicles
	k	Vehicle capacity (maximum number of orders per vehicle)
	y_d	Order delivery cost per unit distance
	y_v	Salary cost of delivery person per unit time
	v	Speed of delivery vehicle
	U_d	Utilization of delivery vehicles

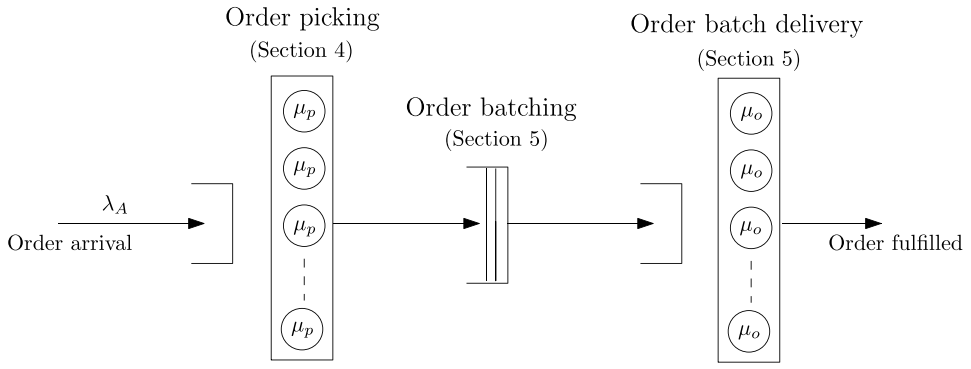


Fig. 2. Open queuing network for the order fulfillment process.

where S is the number of delivery vehicles. The expected waiting time of an order in the delivery queue is given by Eq. (5) (based on Sakasegawa (1977)),

$$W_q = \left[\frac{U_d^{\sqrt{2(S+1)}-1}}{(1-U_d)S}\frac{c_{pb}^2 + c_d^2}{2K}\right]\mu_o^{-1}, \tag{5}$$

where $U_d = \frac{\lambda\mu_o^{-1}}{SK}$ is the utilization of delivery station. The coefficient of variation of waiting time in the order delivery queue is as follows (Zhao et al., 2014):

$$c_q = \sqrt{1 + \frac{4\left(1 - P(T_q > 0)(\psi_q + 2)\right)}{3P(T_q > 0)(\psi_q + 1)}}, \tag{6}$$

where $\psi_q = \frac{1}{c_o^2}$, T_q is the random variable denoting the time an order waits in the order delivery queue, and $P(T_q > 0)$ is the probability that an order waits in the order picking queue. The expression for $P(T_q > 0)$

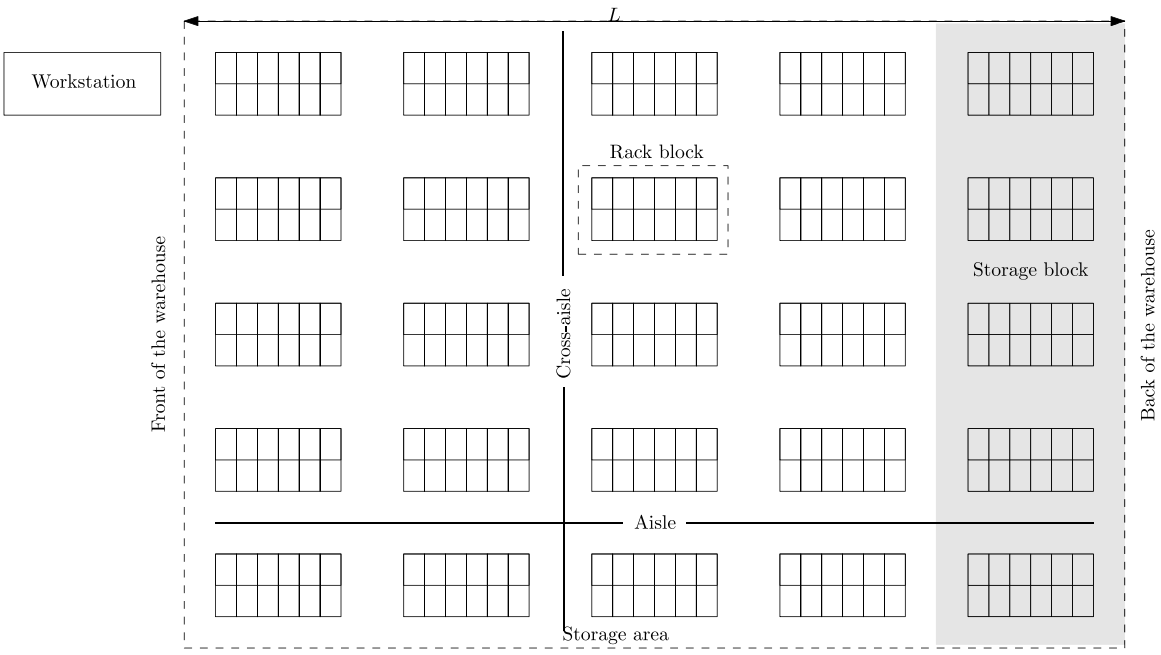


Fig. 3. Warehouse layout.

is given by Zhao et al. (2014),

$$P(T_q > 0) = \frac{r^{\psi_q} c_o(1 - c_{pb}) + \left(\frac{\lambda}{K\mu'_q}\right)(1 - c_o)c_{pb}}{1 - c_o c_{pb}}, \tag{7}$$

where $\mu'_q = \frac{\lambda/K}{(S\mu_p - \lambda/K)W_{qM/M/S}}$. $W_{qM/M/S}$ is the waiting time in $M/M/S$ order delivery queue and r is determined by the real root (< 1) of the equation (Zhao et al., 2014),

$$\mu'_q r^{\psi_q + 1} - (\psi_q \lambda/K + \mu'_q)r + \psi_q \lambda/K = 0$$

Once the batch is assigned to the delivery vehicle, the delivery vehicle takes μ_d^{-1} time on average to deliver a customer order. The estimation of mean (μ_d^{-1}) and variance (σ_d^2) of order delivery time is discussed in detail in Section 4.

4. Analysis of order fulfillment processes

In this section, we estimate the mean and variance of the order picking time, order batching time for delivery, and order delivery time.

4.1. Order picking time

In a manual picker-to-parts order picking system, the picker starts from the workstation with an empty pick cart, retrieves all the line items of an order from the storage area and delivers the cart to the workstation (where picked orders are batched for delivery). The picker processes only one order at a time. Fig. 3 shows the storage area which consists of multiple aisles and cross-aisles (adapted from Roodbergen et al. (2008) and Lamballais et al. (2017)). Such a layout can be found in many warehouses, including online grocery fulfillment centers. The aisles contain racks with items on both sides. The cross-aisles run perpendicular to the aisles and are used by the picker to move from one aisle to another.

The group of *rack blocks* between two adjacent cross-aisles is called a *storage block*. The picker movement is assumed to be bi-directional, i.e., the picker can move in both the direction within the aisles and cross-aisles. We assume aisles are sufficiently wide to prevent picker blocking.

The mean (T_e) and variance (V_e) of order picking distance are estimated in Online Appendix A. Let p_m be the probability that an order consists of m line items and M be the random variable that denotes the number of line items in an order. The mean of the order picking time is the sum of the expected time to pick the line items from the shelf and the expected travel time to retrieve all the line items and deliver them to the workstation. Similarly, the variance of the order picking time is the sum of the variance of the time to pick the line items from the shelf and the variance of the travel time to retrieve the line items and deliver them to the workstation. Then, the expressions for the mean and variance of the order picking time are given as follows:

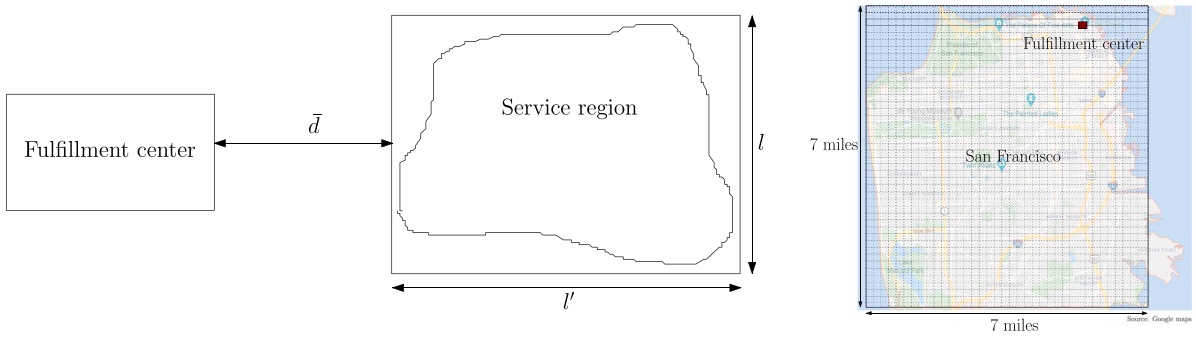
$$\begin{aligned}\mu_p^{-1} &= \frac{E[D]}{u} + E[M]\mu_s^{-1} = \frac{\sum_{m=1}^{M_0} E[D \mid M = m]p_m}{u} + E[M]\mu_s^{-1} \\ &= \frac{\sum_{m=1}^{M_0} T_e(n_a, n_b, m)p_m}{u} + E[M]\mu_s^{-1} \end{aligned} \tag{8}$$

$$\begin{aligned}\sigma_p^2 &= \frac{1}{u^2}V[D] + E[M]\sigma_s^2 = \frac{1}{u^2}\left(V[E[D \mid M]] + E[V[D \mid M]]\right) + E[M]\sigma_s^2 \\ &= \frac{1}{u^2}\left(V[T_e(M)] + E[V_e(M)]\right) + E[M]\sigma_s^2, \end{aligned} \tag{9}$$

where D is the order picking distance. $T_e(n_a, n_b, M)$ (or $T_e(M)$) and $V_e(n_a, n_b, M)$ (or $V_e(M)$) are the expectation and variance (respectively) of the distance traveled by the picker when the order contains M line items and the storage area has n_a number of aisles and n_b number of storage blocks. $E[M]$ is the expected number of line items in an order. The detailed calculations for the expressions $T_e(n_a, n_b, M)$, and $V_e(n_a, n_b, M)$ can be found in Appendix A.

4.2. Order batching time for delivery

The variability in the order batching time for delivery affects the order fulfillment time. We estimate the mean and variance of the batching time for a general order arrival distribution. Let $t = 1/\lambda$ be the expected inter-arrival time between two consecutive orders and σ_{pb}^2 be the variance of the inter-arrival time. Let X be the random variable representing the order waiting time in the batching queue, Y the random variable representing the number of orders waiting in the batching queue, and Z_i be the random variable that denotes the inter-arrival time between order i and $i + 1$. Suppose the random variable Z_i has mean t and variance σ_{pb}^2. Then, $[X \mid Y = i]$ denotes the order



(a) Linking fulfillment center with the delivery service zone

(b) Modeling service region as a rectangular grid

Fig. 4. Integrating fulfillment center with delivery region.

waiting time in the batching queue when the arriving order sees i orders waiting in the queue. Also, $[X \mid Y = 0] = Z_1 + Z_2 + \cdots + Z_{K-1}$. Then, the mean and variance of order waiting time in the batching queue when the arriving order sees i orders in the batching queue are given by,

$$E[X \mid Y = i] = E[Z_1] + \cdots + E[Z_{K-i-1}] = (K-1-i)t$$
$$V[X \mid Y = i] = V[Z_1] + \cdots + V[Z_{K-i-1}] = (K-1-i)\sigma_{pb}^2,$$

where $i \in \{0, 1, 2, \ldots, K-1\}$. Then, the mean, variance, and squared coefficient of variation of the order waiting time in the batching queue are as follows:

$$\begin{aligned}\text{Mean, } \mu_b^{-1} &= E[X] = E[E[X \mid Y]] \\ &= \frac{0+1+2+\cdots+(K-2)+(K-1)}{K}t \\ \implies \mu_b^{-1} &= \frac{(K-1)t}{2}, \end{aligned} \tag{10}$$

$$\begin{aligned}\text{Variance, } \sigma_b^2 &= V[X] \\ &= E[V[X \mid Y]] + V[E[X \mid Y]] \\ &= E[V[X \mid Y]] + E[E[X \mid Y]]^2 - [E[E[X \mid Y]]]^2 \\ &= \frac{\sigma_{pb}^2(0+1+\cdots+(K-1))}{K} + \frac{t^2(0^2+1^2+\cdots+(K-1)^2)}{K} \\ &\quad - \left[\frac{(K-1)t}{2}\right]^2 \\ \implies \sigma_b^2 &= \frac{(K-1)\sigma_{pb}^2}{2} + \frac{t^2(K^2-1)}{12}, \text{and} \end{aligned} \tag{11}$$

$$c_b^2 = \sigma_b^2\mu_b^2 = \frac{2c_{pb}^2}{(K-1)t^2} + \frac{K^2-1}{3(K-1)^2}. \tag{12}$$

4.3. Order delivery time

Suppose the fulfillment center caters to the demand of a service region with arbitrarily-shaped service region (see Fig. 4(a)). Customer orders arrive at the fulfillment center and the fulfillment center delivers these orders to the delivery point or the customer's location. The delivery points are scattered uniformly in the service region.

The expected distance traveled to deliver the order batch of size K consists of two components — the distance required to reach the periphery of the service region called 'line-haul distance' ($\bar{d}$) and the distance of the optimal traveling salesman tour to cover the K random locations. We assume that the road network in the service region has a rectangular grid structure. The traveling salesman problem (TSP) tour length is estimated in the same way as the order picking distance. However, now it is a delivery vehicle traveling to the customer locations to deliver the orders, instead of an order picker traveling to the locations of line items. To estimate the TSP tour distance, we use the number of aisles and blocks of the road network in the rectangular service region grid structure. Let n_a^s and n_b^s be the number of aisles and blocks in the rectangular grid structure. Then, we can now use Eqs. (8) and (9) to estimate the mean and variance of the time taken by a delivery vehicle to complete the TSP tour. Let δ_d be the expected distance traveled to deliver a batch of orders and to return to the fulfillment center, then δ_d equals $T_e(n_a^s, n_b^s, K) + 2\bar{d}$, where $\bar{d}$ is the mean travel distance to a first customer in the service region (see Appendix A for the expressions of $T_e(n_a^s, n_b^s, K)$ and $V_e(n_a^s, n_b^s, K)$). A certain stop time is required at each delivery point. The mean and variance of each stop time per order are μ_t^{-1} and σ_t^2. The mean and variance of loading a batch of orders on the delivery vehicle are μ_l^{-1} and σ_l^2. Then, the mean and variance of the time taken by the delivery vehicle to deliver a batch of orders are given as follows:

$$\mu_o^{-1} = \frac{\delta_d}{v} + K\mu_t^{-1} + \mu_l^{-1}, \tag{13}$$

$$\sigma_o^2 = \frac{1}{v^2}\left[V_e(n_a^s, n_b^s, K) + 2V_d\right] + K\sigma_t^2 + \sigma_l^2, \tag{14}$$

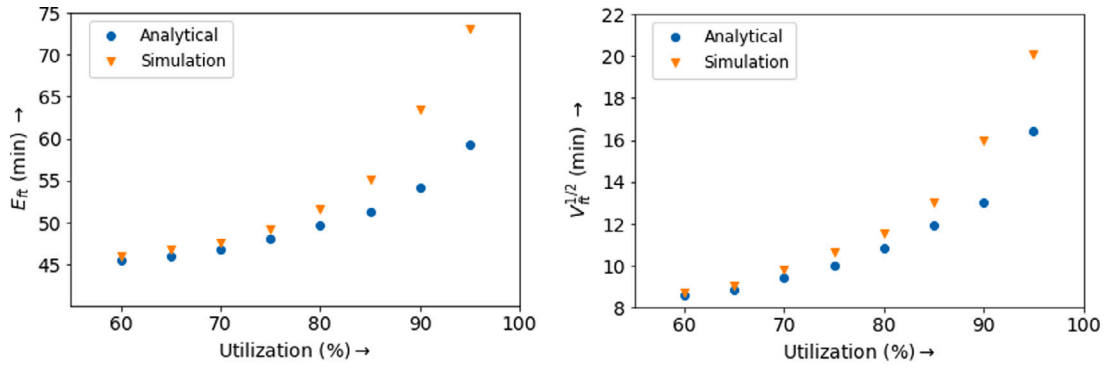
where V_d is the variance of the travel distance by the delivery vehicle to reach a first customer location in the service region. The mean of the time between two consecutive customer order deliveries is $\frac{T_e(n_a^s, n_b^s, K)}{Kv}$ and the variance of the time between two consecutive customer order deliveries is $\frac{V_e(n_a^s, n_b^s, K)}{Kv^2}$. The mean and variance of the order delivery time are given as follows:

$$\mu_d^{-1} = \frac{1}{v}\left[\frac{T_e(n_a^s, n_b^s, K) + 2\bar{d} + \mu_l^{-1}}{K}\right] + \mu_t^{-1}, \tag{15}$$

$$\sigma_d^2 = \frac{1}{v^2}\left[\frac{[V(n_a^s, n_b^s, K) + 2V_d + \sigma_l^2]}{K}\right] + \sigma_t^2. \tag{16}$$

In order to estimate the quality of the travel time estimates, a detailed simulation validation of the analytical model is carried out. The details can be found in Online Appendix B. The absolute difference in waiting times is less than one minute. The average percentage errors for the mean and standard deviation of order fulfillment throughput times (order picking, batching, and delivery) are less than 5% and 15%, respectively (see Table B2 in Appendix B) .

Heavy traffic: The quality of Whitt's method generally depends e.g. on the utilization rate. Roy et al. (2020), for example, show that at very high utilization, the deviations can be large. We, therefore, compare the performance of our integrated analytical model with a simulation model at different utilization levels (see Fig. 5), where we used the simulation validation model explained in Appendix B. The results show that the average difference in mean and variance of order fulfillment time at low (60%) and very high utilization levels (95%) is 1.1% and 23.5% and 1.9% and 22.6%, respectively. However, we note that at a 90% or 95% delivery probability (DP) level (i.e., delivery reliability) and 60 min delivery lead time threshold, the picker and rider utilization



(a) Utilization vs E_{ft} (b) Utilization vs $\sqrt{V_{ft}}$

Fig. 5. Comparison of analytical model with simulation model at low and high utilization levels.

lies in between 60% and 65% (for an arrival rate range of 10–35 orders/hr). Hence, the estimate errors introduced due to analytical approximations are acceptable.

5. Optimization model and solution methodology

In this section, we build an optimization model to minimize the expected order fulfillment cost subject to DP constraint. As decision variables, we use the number of pickers, the size of the order batch for delivery, and the number of delivery vehicles. The DP is the index used to measure the performance of the order fulfillment process (Garg et al., 2006). DP is the delivery reliability, i.e. the probability of fulfilling the customer order within the specified delivery time window. We minimize the expected variable order fulfillment cost, which is the sum of the order picking cost and order delivery cost. The decision variables are the number of pickers (N), order batch size for delivery (K), and the number of delivery vehicles (S). Delivery cost can be divided into distance related cost and delivery person salary cost.

Let the order arrival rate be λ orders/hr. The cost of delivering an order is $\frac{y_d \delta_d}{K}$, where y_d is the order delivery cost per unit distance. Let y_p be the picking cost/hour per picker, then the picking cost per hour of N pickers is $y_p N$. The picking cost per order is $\frac{y_p N}{\lambda}$. Similarly, we calculate the salary cost of the delivery person per order. The delivery person's salary cost per order is $\frac{y_v S}{\lambda}$, where y_v is the salary cost of the delivery person per time unit. In the optimization model, we assume that the layout of the storage area, the delivery service region (size and layout), and the order arrival rate (as well as the associated CV) are known a priori. Then, the model minimizing the expected order fulfillment cost subject to a DP constraint is as follows.

$$\min_{N,K,S} C = \frac{y_p N}{\lambda} + \frac{y_d \delta_d}{K} + \frac{y_v S}{\lambda}, \tag{17}$$

subject to

$$DP := P(X_{ft} \le D_u) \ge d, \tag{18}$$

We approximate the order fulfillment time distribution using a normal distribution, since the mean of order fulfillment time in practice is sufficiently large — typically around 0.75 h. Fig. 6 shows the empirical distribution of order fulfillment times obtained from an external party (a quick-commerce grocery deliver company from India), with an average of 50 min. The inequality corresponding to DP is given in Eq. (18), where X_{ft} is the random variable denoting order fulfillment time. Further D_u is the target for due time and d is the target DP that this set by the company. For example, if the target due time for order fulfillment is 60 min and the company strives to deliver at least 90% of the orders within the due time, then D_u is 60 min and d is 90%.

The DP constraint ensures that customers receive their orders within the time limit (D_u). From Eq. (18), we can observe that a low variance of order fulfillment time results in a higher DP value. It means that the DP constraint ensures an optimal solution with a low variance of order fulfillment time. However, the cost-optimal solution that satisfies the DP constraint might be different from the cost-optimal solution without considering this constraint. Hence, achieving lower variance might also impact the expected order fulfillment time and cost. Section 7 investigates this impact.

The expected order fulfillment time is the sum of the expected time an order waits at the order picking station, expected order picking time, expected waiting time to form the batch, expected time an order waits in the delivery queue, and expected time to deliver an order. The expression for E_{ft} can be found from Eqs. (1), (5), (8), (10), and (15) as follows:

$$E_{ft} = \left[\frac{U_p^{\sqrt{2(N+1)}-1}}{(1-U_p)N}\frac{c_a^2+c_p^2}{2}\right]\mu_p^{-1} + \mu_p^{-1} + \frac{K-1}{2\lambda} + \left[\frac{U_d^{\sqrt{2(S+1)}-1}}{(1-U_d)S}\frac{c_{pb}^2+c_d^2}{2K}\right]\mu_o^{-1} + \mu_d^{-1}. \tag{19}$$

To determine the variance of order fulfillment time, we assume that the order fulfillment processes are independent of each other. Thus, we can compute the variance of order fulfillment time (V_{ft}) by the sum of the variances of waiting time in the order picking queue, order picking time, order batching time for delivery, waiting time in the order delivery queue, and order delivery time. The expression for V_{ft} can be obtained from Eqs. (2), (6), (9), (11), and (16) as follows:

$$\begin{aligned} V_{ft} &= \sigma_r^2 + \sigma_p^2 + \sigma_b^2 + \sigma_q^2 + \sigma_d^2 \\ &= \left[\sqrt{1 + \frac{4\left(1 - P(T_r > 0)(\psi_r + 2)\right)}{3P(T_r > 0)(\psi_r + 1)}}\right]^2 \left[\left(\frac{U_p^{\sqrt{2(N+1)}-1}}{(1-U_p)N}\frac{c_a^2+c_p^2}{2}\right)\mu_p^{-1}\right]^2 \\ &\quad + \sigma_p^2 + \frac{(K-1)\sigma_{pb}^2}{2} \\ &\quad + \frac{t^2(K^2-1)}{12} + \left[\sqrt{1 + \frac{4\left(1 - P(T_q > 0)(\psi_q + 2)\right)}{3P(T_q > 0)(\psi_q + 1)}}\right]^2 \\ &\quad \times \left[\left(\frac{U_d^{\sqrt{2(S+1)}-1}}{(1-U_d)S}\frac{c_{pb}^2+c_d^2}{2K}\right)\mu_o^{-1}\right]^2 + \sigma_d^2. \end{aligned} \tag{20}$$

To solve this optimization model, we use a linear search algorithm (see Appendix D for the pseudo-code of the algorithm), we provide the upper and lower bounds of the decision variables and then search for the optimal solution within those bounds. We use the fact that the lower bound for batch size is one and the upper bounds of the batch size is k (delivery vehicle capacity). The lower bound of N is calculated using the queue stability condition for the order picking station, which leads to $N_{lb} = \left\lfloor \lambda \mu_p^{-1} \right\rfloor + 1$. The upper bound of N is calculated using the

square root staffing rule. Under the square root staffing rule, the value of the upper bound of the number of pickers N is given by Gans et al. (2003), Whitt (1992), and Borst et al. (2004),

$$N_{ub} = \left\lfloor R_p + \alpha\sqrt{R_p} \right\rfloor, \tag{21}$$

where $R_p = \lambda\mu_p^{-1}$ and α is approximated using Eq. (22) provided by Whitt (2004).

$$1 - SL = \frac{1}{1 + \frac{\alpha\Phi(\alpha)}{\phi(\alpha)}}, \tag{22}$$

where Φ and ϕ are the distribution and density functions of the standard normal distribution, respectively. The service level (SL) is the probability that an order does not wait in the queue. For example, if $SL = 0.9$, 90% of the orders do not wait in the queue. The value of α for $SL = 0.9$ is 1.28. To estimate the lower and upper bound of the number of delivery vehicles required, we are using the lower and upper bounds of the batch size. We define the utilization of delivery vehicles as the probability that the delivery vehicles are in use to deliver customer orders. The utilization of the delivery vehicles decreases as we increase the batch size, meaning that the number of delivery vehicles required to deliver the orders does not increase as we increase the batch. Thus, we use the lower bound of the delivery batch size to determine the upper bound of the number of delivery vehicles which is given by,

$$S_{ub} = \left\lfloor \frac{\lambda}{\mu_o\big|_{K=1}} + \alpha\sqrt{\frac{\lambda}{\mu_o\big|_{K=1}}} \right\rfloor. \tag{23}$$

We use the upper bound of the batch size to determine the lower bound of the number of delivery vehicles. The queue stability condition gives the minimum number of delivery vehicles required for a given batch size. Combined, this gives the following lower bound of the number of delivery vehicles:

$$S_{lb} = \left\lfloor \frac{\lambda}{k\mu_o\big|_{K=k}} \right\rfloor + 1. \tag{24}$$

Hence, the lower and upper bounds of the number of delivery vehicles are $S_{lb} = \left\lfloor \frac{\lambda}{k\mu_o\big|_{K=k}} \right\rfloor + 1$ and $S_{ub} = \left\lfloor \frac{\lambda}{\mu_o\big|_{K=1}} + \alpha\sqrt{\frac{\lambda}{\mu_o\big|_{K=1}}} \right\rfloor$, respectively. Now, we determine the optimal number of pickers, order batch size for delivery, and the number of delivery vehicles. First, we determine the lower and upper bounds of the decision variables. Then, we calculate the expected order fulfillment cost for all the feasible triplets of the number of pickers, order batch size, and the number of delivery vehicles. Then, we choose the solution with the minimum expected order fulfillment cost as the optimal solution to the optimization problem.

6. External-party model validation

We show the validation of the performance measures obtained from the analytical model using a detailed simulation model (Online Appendix B). However, in this section, we present an application of our analytical model to a real world quick-commerce case. We partner with a hyperlocal grocery order delivery service company that fulfills quick orders with a due time of one hour. We use our analytical model to determine the optimal operational parameters (number of pickers, delivery batch size, and number of delivery vehicles).

This company recently started a quick service for their grocery segment in which they promise to fulfill the customer orders within one hour. The data from the company included 2148 quick orders received during the week, 7 Feb to 13 Feb 2022. The company plans handling and delivery capacity using a two hour slot that starts from 6 am to 10 pm. This data-set gives us information regarding the start and end time-stamps of every process, such as order creation time, pick list creation time, pick start time, pick end time, batch end time, rider arrival at customer location time, and order receiving time. We use the time-stamp to determine the actual values of mean and variance of the order fulfillment process times. We use the unique picker ID and batch ID information to determine the number of pickers and delivery batch size. The data regarding the requested third-party delivery capacity corresponding to all slots is provided by the company. The operational cost parameters are adopted from Table 5. We use these data in our analytical model and observe the actual vs estimated output performance measures.

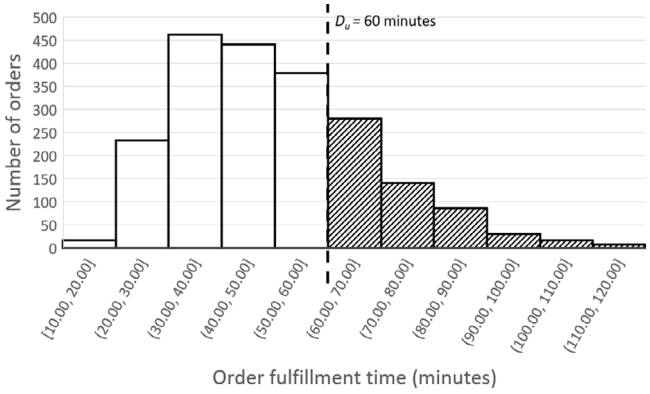


Fig. 6. Actual order fulfillment time (in minutes).

Table 2 shows the actual utilization of pickers and delivery vehicles. The results of the validation exercise for batching time and waiting time in the order picking and delivery queues are given in Table 2. As before, W_r, $1/\mu_b$, and W_q are the estimated expected waiting time in the order picking, batching, and delivery queues obtained from our analytical model, whereas W_r^e, $1/\mu_b^e$, and W_q^e are the expected waiting time in the order picking, batching, and delivery queues from the external party. The average picker waiting times estimated from our model (W_r) and external data (W_r^e) across all slots are 0.646 min and 0.801 min, respectively. Likewise, the average wait across all slots for batch formation estimated from our model ($1/\mu_b$) is 7.168 min, and from external data ($1/\mu_b^e$), it is 7.865 min, respectively. The results show that the estimated values of waiting time in order picking and batching queues reasonably match with the actual measures for practical purposes. The waiting time for riders is almost negligible in the external party data because the delivery vehicles (riders) are almost always stationed at the dark store and have a low utilization (as seen in Table 2). Our model also estimates a zero wait time for the delivery vehicle (rider) availability.

The actual average travel time for delivery is approximately one hour. The reason for the low waiting time in the delivery queue is the low utilization of riders, see Table 2, hence estimated expected waiting time in the delivery queue is close to zero. We now use our model to find out the level of the different parameters (number of pickers, delivery batch size, and number of delivery vehicles) to meet the desired service level at minimum cost for the company's quick orders.

We determine the optimal setting for the operational parameters, $\{N^*, K^*, S^*\}$, from our model and compare them with the settings currently in use by the company. Fig. 6 shows the histogram of actual fulfillment time for the orders. The average order fulfillment time is 50 min, which is shorter than the promised due time of 60 min. From the figure, we can observe that the company can fulfill approximately 72% (1545 orders out of 2148 orders) within the due time of one hour. So, it is important to estimate the variance to determine the attainable service quality and due time for customer orders. To improve the service quality of order fulfillment, we use our model to find the solution that meets their service quality criteria at minimum cost.

Table 2
External validation of expected waiting time in order picking, batching, and delivery queues. Superscript e against a measure, $(.)^e$ denotes the external party's output measure. Both $(W_q)^e$ and $(W_q)^e$ are approximately zero for all slots and therefore not listed.

Slot	λ (orders/hr)	U_p, U_p^e (%)	U_d, U_d^e (%)	W_r^e (min)	W_r (min)	$1/\mu_b^e$ (min)	$1/\mu_b$ (min)
6 am–8 am	5	23%	31%	0.710	0.185	9.244	11.34
8 am–10 am	8	38%	23%	0.699	0.337	5.934	7.646
10 am–12 pm	10	36%	22%	0.619	0.408	6.917	6.731
12 pm–2 pm	12	48%	23%	0.638	0.690	7.902	5.944
2 pm–4 pm	10	45%	22%	0.698	0.671	6.753	6.357
4 pm–6 pm	12	53%	23%	1.628	1.530	7.200	6.057
6 pm–8 pm	13	52%	24%	1.018	0.789	9.000	5.567
8 pm–10 pm	8	41%	19%	0.400	0.561	9.967	7.705

Table 3
The company's actual vs. proposed operational parameters and corresponding cost per order.

Slot	Actual									Proposed								
	U_p	U_d	N	K	S	E_{ft} (h)	$\sqrt{V_{ft}}$ (h)	DP	C ($/order)	U_p	U_d	N^*	K^*	S^*	E_{ft} (h)	$\sqrt{V_{ft}}$ (h)	DP	C ($/order)
Slot 1 (6 am–8 am)	23%	31%	5	2	15	0.73	0.28	0.83	3.42	28%	44%	3	1	11	0.48	0.17	0.95	3.12
Slot 2 (8 am–10 am)	38%	23%	5	2	43	0.74	0.26	0.84	5.30	53%	54%	3	1	19	0.54	0.19	0.95	4.00
Slot 3 (10 am–12 am)	36%	22%	5	2	50	0.80	0.28	0.76	5.58	51%	63%	3	1	18	0.48	0.17	0.95	3.88
Slot 4 (12 am–2 pm)	48%	23%	5	2	54	0.79	0.34	0.73	5.80	66%	56%	3	1	23	0.54	0.19	0.95	4.15
Slot 5 (2 pm–4 pm)	45%	22%	5	2	50	0.85	0.36	0.66	5.46	60%	54%	3	1	21	0.55	0.19	0.95	3.91
Slot 6 (4 pm–6 pm)	53%	23%	5	2	58	0.93	0.32	0.58	6.02	72%	47%	3	1	30	0.72	0.27	0.95	4.52
Slot 7 (6 pm–8 pm)	52%	24%	5	2	62	0.97	0.31	0.54	6.27	65%	58%	3	1	27	0.56	0.19	0.95	4.42
Slot 8 (8 pm–10 pm)	41%	19%	5	2	50	0.87	0.27	0.69	5.56	67%	65%	3	1	15	0.46	0.15	0.95	3.71
Average						0.84	0.30	0.71	5.43						0.54	0.19	0.95	3.96

Table 3 presents the company's actual service quality. On average, the company can fulfill 71% of its orders within one hour. Next, our analytical model is used to determine the workforce, delivery batch size, and due time such that 95% of the orders are delivered within one hour at minimum expected order fulfillment cost. Table 3 shows the actual and proposed operational parameters for the company's quick orders. Currently, the company uses an average delivery batch size of two orders which results in a low DP and long lead times. We propose to deliver only one order per delivery trip. This significantly reduces the order fulfillment time and allows shorter due times. This case study shows how our analytical model can be applied to determine optimal operational parameters (number of pickers, delivery batch size, and number of delivery vehicles) which minimizes the expected order fulfillment cost and to set due times for q-commerce. The results show that the company is able to fulfill 95% of its orders within one hour using our model, as compared to 72% using its current operational strategy. This outcome can be achieved with an average cost savings of $1.47 per order (27% cost reduction). However, we show (in Table 8) that even in peak hours the batch size can be lowered to 1, at a lower cost than the current, with 95% of the orders delivered within one hour. Also, the number of pickers and delivery vehicles can be lowered. We have carried out an additional analysis and found that the company's setting with $N = 5$ pickers, a batch size $K = 2$, and about $S = 50$ vehicles is close to optimal when the order arrival rate is almost double in the peak hour than current. Table 3 presents the company's actual service quality. On average, the company can fulfill 71% of its orders within one hour. Next, our analytical model is used to determine the workforce, delivery batch size, and due time such that 95% of the orders are delivered within one hour at minimum expected order fulfillment cost. Table 3 shows the actual and proposed operational parameters for the company's quick orders. Currently, the company uses an average delivery batch size of two orders which results in a low DP and long lead times. We propose to deliver only one order per delivery trip. This significantly reduces the order fulfillment time and allows shorter due times. This case study shows how our analytical model can be applied to determine optimal operational parameters (number of pickers, delivery batch size, and number of delivery vehicles) which minimizes the expected order fulfillment cost and to set due times for q-commerce. The results show that the company is able to fulfill 95% of its orders within one hour using our model, as compared to 72% using its current operational strategy. This outcome can be achieved with an average cost savings of $1.47 per order (27% cost reduction). However, we show (in Table 8) that even in peak hours the batch size can be lowered to 1, at a lower cost than the current, with 95% of the orders delivered within one hour. Also, the number of pickers and delivery vehicles can be lowered. We have carried out an additional analysis and found that the company's setting with $N = 5$ pickers, a batch size $K = 2$, and about $S = 50$ vehicles would be close to optimal when the order arrival rate in the peak hour (6:00–8:00 pm) would double compared to the current arrival rate.

In q-commerce, profitability depends on the promised fulfillment time and customer demand. Customers like a shorter fulfillment time (possibly creating greater demand), but to achieve a shorter fulfillment time costs increase. We develop an econometric model to analyze this trade-off in Appendix E, where we show demand maximizing retailers may decide to offer a very short fulfillment time. However, this strategy may not result in the maximum profit which may be achieved by offering a longer promised fulfillment time.

7. Results and managerial insights

This section presents the numerical results obtained with the analytical model proposed in Section 5. We use customer order data obtained from Instacart in the numerical experiments.

7.1. Dataset information

The parameters used in the numerical experiments can be grouped into three categories: customer order-related information, order delivery process information, and order picking process information.

The Instacart Online Grocery Shopping Dataset (2017) includes about 1.4 million customer orders in the San Francisco area. Instacart is a speedy (30–45 min delivery time) online grocery delivery company in the US and Canada that partners with more than 1100 retailers spanning 80,000 stores. From these data, we obtain order characteristics such as the number of line items per order and order arrival rate. We use an order arrival rate range from 10 to 35 orders per hour, based on the actual order arrival rates for the quick-commerce retailer of Section 6. Table 4 summarizes the parameters for the numerical experiments.

In the order delivery process, we assume that the customer locations are uniformly distributed in the San Francisco service region

Table 4
Customer order information.

Name	Parameter	Value
Order arrival rate (orders/hr)	λ	10–35
Coefficient of variation of order inter-arrival time	c_a	{0.5, 1, 1.5, 2,2. 5,3}
Probability of an order with m line items	p_m	See Fig. B2 in Appendix B
Maximum number of line items in an order	M_0	80
Due time (min)	D_u	{30, 45, 60}
Threshold for DP	d	{0.9, 0.925, 0.95}

(see Fig. 4(b)). We use the parameter values from the study by Quak and De Koster (2007): the order delivery cost per kilometer (y_d) is \$0.3/km, the salary cost of delivery personnel (y_v) is \$20/h, and the average speed of the delivery vehicle (v) is 25 km/h. Table 5 presents the delivery parameters. The parameters of the order picking process are based on the parameters of the e-commerce fulfillment center presented by Lamballais et al. (2017). Table 5 summarizes the order pick parameters.

7.2. Results

To obtain numerical results, we use the following base parameters in this section: $k = 25$ orders, $N_{lb} = 5$, $N_{ub} = 9$, $S_{lb} = 3$, S_{ub}=13, and $d = 0.9$. The storage area in the fulfillment center has 14 aisles and 13 blocks and the parameters related to the service region are as described in Section 7.1. The base values for the parameters D_u and c_a are taken from the range given in Table 4, i.e. $D_u = 60$ and $c_a = 1$.

Effect of staffing and batching decisions on order fulfillment measures

We first analyze the impact of order fulfillment resources (number of pickers and delivery vehicles) and delivery batch size on expected order fulfillment time, expected order fulfillment cost, and DP. To analyze the effect of order fulfillment resources, we use the percentage change in the value of the performance measures. The percentage change is defined as $\frac{P(current)-P(optimal)}{P(optimal)} \times 100$, where $P(current)$ is the value of performance measure when the number of resources takes a fixed value, and $P(optimal)$ is the value of the performance measure when the number of resources is based on the outcome of the optimization model. The value of performance measures at the optimal (base line) solution, obtained from model (17)–(18), are: C = \$12.3 per order, E_{ft} = 45.5 min, $\sqrt{V_{ft}}$ = 8.5 min, $N = 6$, $K = 6$, S= 7, and $DP = 0.9$.

Effect of the number of pickers: We find that most measures, except order fulfillment cost, change only marginally when the number of pickers increases. Although the number of pickers affects the downstream variability of job inter-arrival times, the magnitude is quite small. Increasing the number of pickers increases upstream picker cost and hence overall order fulfillment cost.

Effect of the number of delivery vehicles: Fig. 7 presents a comparison between the performance measures for a varying number of delivery vehicles and the performance measures when the number of delivery vehicles is optimal (S^*=7). The y-axis of the graph shows the percentage change (increase or decrease)when the outcome of the optimization model for a given number of delivery vehicles is compared with the outcome with the optimal number of delivery vehicles.

The results show that the expected order fulfillment cost and DP decreases if there are fewer delivery vehicles than the optimal quantity. However, the mean and variance of order fulfillment time and delivery batch size increase. A large delivery batch size results in a higher variance of order batching time before delivery, so the variance of order fulfillment time increases which leads to a lower DP. Also, using fewer delivery vehicles leads to lower expected order fulfillment cost. The results show that using more delivery vehicles than the optimal quantity leads to a very small change in DP, and variance of order

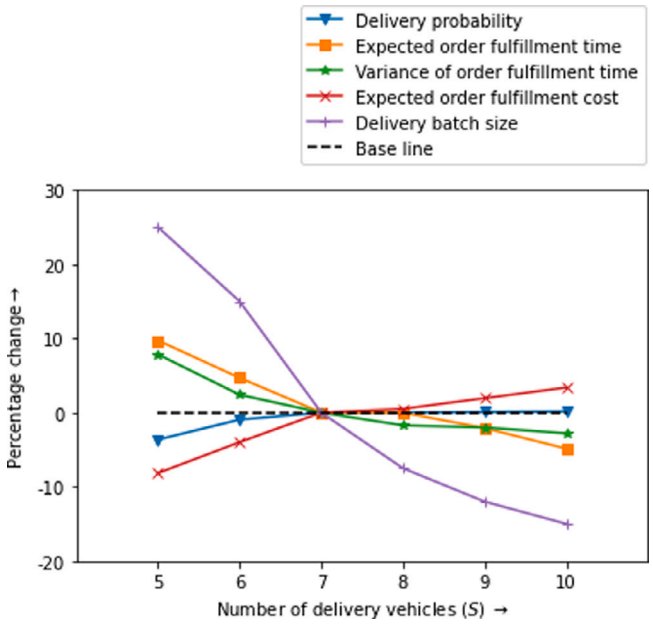


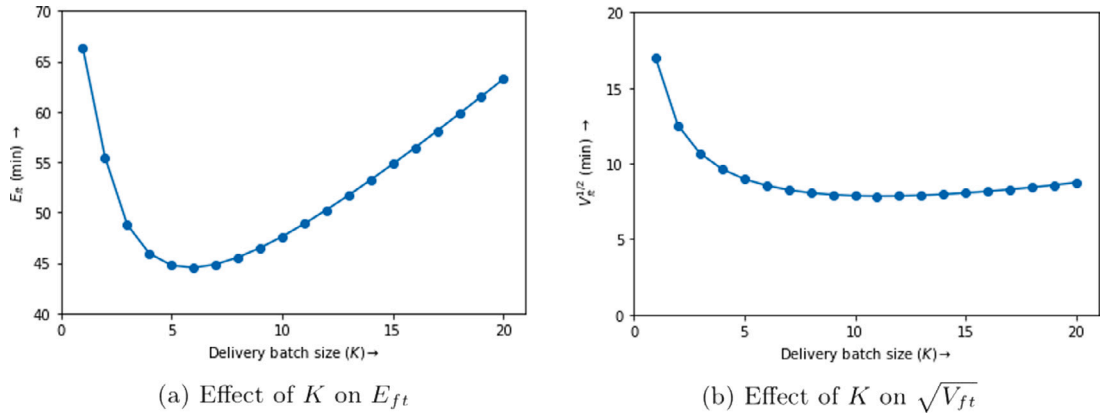
Fig. 7. Percentage change for a given number of delivery vehicles (the optimal number of delivery vehicles is 6 in the base line model).

fulfillment time. The order fulfillment cost increases and expected order fulfillment time decreases as the number of delivery vehicles increase. A large number of delivery vehicles results in a short waiting time at the order delivery queue, reducing the expected order fulfillment time. We find that delivery batch size decreases as we increase the number of delivery vehicles. Apparently, the smaller batch size does not increase the order delivery time since a large number of delivery vehicles are available.

Effect of delivery batch size: Fig. 8 shows the effect of batch size on the mean and standard deviation of order fulfillment time. The parameters related to the storage area and service region are defined as in Section 7.1. In Fig. 8(a), the X-axis shows the batch size used in the optimization model (i.e., N and S optimized for a given K), and the Y-axis denotes the expected order fulfillment time obtained from the solution of optimization model (17)–(18) for a fixed batch size K. Similarly, in Fig. 8(b), the Y-axis indicates the standard deviation obtained from the solution of the optimization model with a fixed batch size K. As expected, the expected order fulfillment time decreases and then increases in a convex fashion with larger batch sizes. For large batch sizes, the increase in order batch completion time dominates the batch waiting time for delivery, resulting in a near-linear increase in E_{ft}. The standard deviation of the order fulfillment time also decreases and then increases with larger batch sizes. This convex-like trend in the standard deviation can be explained by the pattern of the variance terms V_d in Eq. (13), which is the variance of the distance traveled by the delivery vehicle to reach a customer location in the service region. While the V_d term decreases with batch size, the other terms increase with batch size. Hence, we obtain a convex-like graph for the relationship between batch size and standard deviation of the order fulfillment time. This result has implications on how the fulfillment targets should be set. For example, assume the fulfillment center wishes to deliver the order in 45 min and the delivery batch size is 1. However, the expected order fulfillment time for delivery batch size 1 is 66.3 min, so the fulfillment center may find it difficult to deliver an order within 45 min. Thus, the fulfillment center has to redefine its order fulfillment targets depending on the delivery batch size. To explore the effect of batch size further, we evaluate the impact of dynamic batching in Appendix F. Our results show that dynamic batching has a significant impact when the delivery vehicle utilization is high. However, dynamic

Table 5
The values of order fulfillment process parameters.

Process	Name	Parameter	Value
Order picking process	Mean and standard deviation of the time to pick a line item in the storage area (sec)	μ_s^{-1}, σ_s	10, 5
	Picking cost per hour per picker ($/h)	y_p	20
	Picker speed (m/sec)	u	1
Order delivery process	Vehicle capacity (number of orders)	k	{1, 10, 20, 30, 40, 50}
	Mean and standard deviation of stop time per order (min)	μ_t^{-1}, σ_t	3, 2
	Mean and standard deviation of batch loading time on the delivery vehicle (min)	μ_l^{-1}, σ_l	3, 1
	Delivery cost per kilometer ($/km)	y_d	0.3
	Salary cost of delivery person per hour ($/h)	y_v	20
	Speed of the delivery vehicle (km/h)	v	25



(a) Effect of K on E_{ft} (b) Effect of K on $\sqrt{V_{ft}}$

Fig. 8. Effect of fixed delivery batch size on the mean and standard deviation of order fulfillment time.

batching improves the order fulfillment time only marginally as the variability in order inter-arrival time increases.

Effect of service quality constraints on performance measures

In this section, we explain the impact of the service quality constraint (18) (DP) on the performance measures: expected order fulfillment time and variance of order fulfillment time. We compare the performance measures using two scenarios: (1) by including the DP constraint which we refer to as 'Including Service Constraint' (ISC) (2) by excluding the DP constraint which we refer to as 'Excluding Service Constraint' (ESC). This comparison allows us to understand the effect of DP constraints on the various performance measures. Fig. 9 shows the effect of order arrival rate on various performance measures when the DP constraint is present or absent in the optimization model.

Figs. 9(a) and 9(b) show how the order arrival rates affect the mean and standard deviation of the order fulfillment time. These figures show that the ESC model has a higher expected order fulfillment time (44.1% on average) and standard deviation of order fulfillment time (18.6% on average) than the ISC model. In this ESC model, for $\lambda = 25$ orders per hour, N^*, K^*, and S^* take the values of 6, 25, and 3, respectively, whereas in the ISC model, N^*, K^*, and S^* take the values of 6, 6, and 7, respectively. A larger optimal batch size in the ESC model compared to the ISC model (25 vs 6) results in a shorter delivery lead time per customer. In the ESC model, the batching time is 28.8 min and the order delivery time is 6.7 min for $\lambda = 25$ orders per hour, compared to the ISC model where these times are 6 and 16.8 minutes, respectively. The pick time in both models is unaffected as it primarily depends on the warehouse layout. So overall, the ESC model shows a time increase of 12.7 min in comparison to the ISC model. Measured over all arrival rates, it appears that the optimal batch size in the ISC model is considerably lower than in the ESC model. So there is a trade-off between batching time and delivery time.

Cost-quality trade-off

There is a trade-off between order fulfillment cost and the service quality of the order fulfillment process (measured by DP). Fig. 10 shows the cost-quality trade-off for the order fulfillment process. Fig. 10(a) shows the DP and order fulfillment cost trade-off. Fig. 10(b) shows the trade-offs between the order batch size for delivery and DP. Fig. 10(c) depicts these optimal decision-quality trade-offs for the number of delivery vehicles.

We define a solution for the order fulfillment problem with a triplet $\{N, K, S\}$. The most cost-effective solution (i.e. neglecting the DP constraint) is {6, 25, 3}. This solution gives expected order fulfillment cost of $8.1 per order with a DP of 57.6%. If the fulfillment center wants to improve DP, it needs to invest in order fulfillment resources, leading to higher costs. By increasing the number of delivery vehicles by two and reducing the delivery batch size by ten (the new solution is {6, 15, 5}), the fulfillment center can achieve a DP of 79.6% in comparison to 57.6% in the cost-effective case. However, this increases the order fulfillment cost by 16.5% (from $8.1 to $9.4 per order). Therefore, the fulfillment center needs to analyze these cost-quality trade-offs for efficient resource planning.

The value of integration

To determine the optimal number of pickers, delivery batch size, and number of delivery vehicles, an integrated decision support tool is required. It is easy to verify that the delivery batch size K affects the number of delivery vehicles S required for the order fulfillment. So the batch size and number of delivery vehicles decisions should be made while considering integrated batching and delivery processes. Table 6 shows the impact of the picker utilization (U_p) on σ_r, σ_b, σ_d and E_{ft}. The results show that an increase in picker utilization leads to an increase in σ_r and a decrease in σ_b which is a function of c_{pb} (see Eq. (4)). As we increase the value of U_p from 50% to 95%, the value of c_{pb} decreases from 0.9 to 0.6 which results in a decrease in

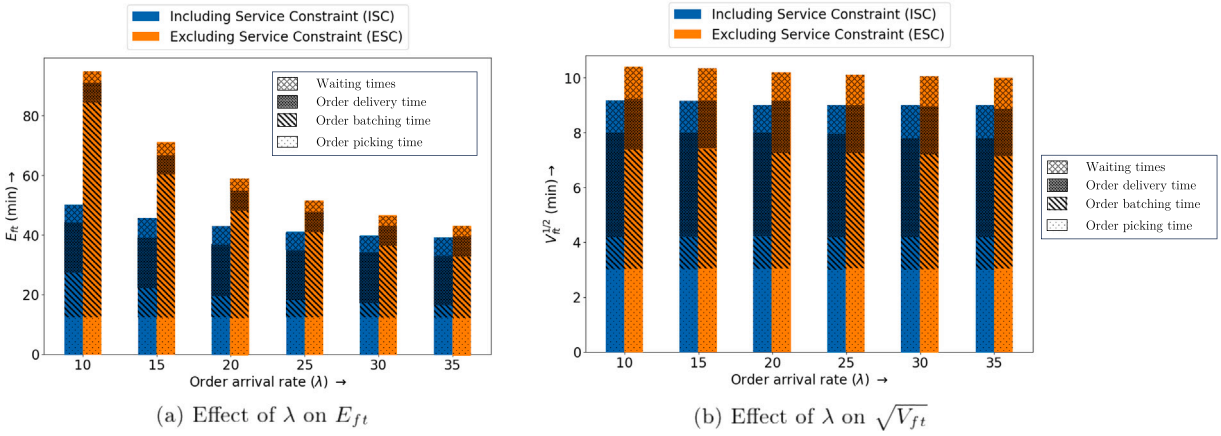


Fig. 9. Effect of service quality constraint on the mean and variance of order fulfillment time.

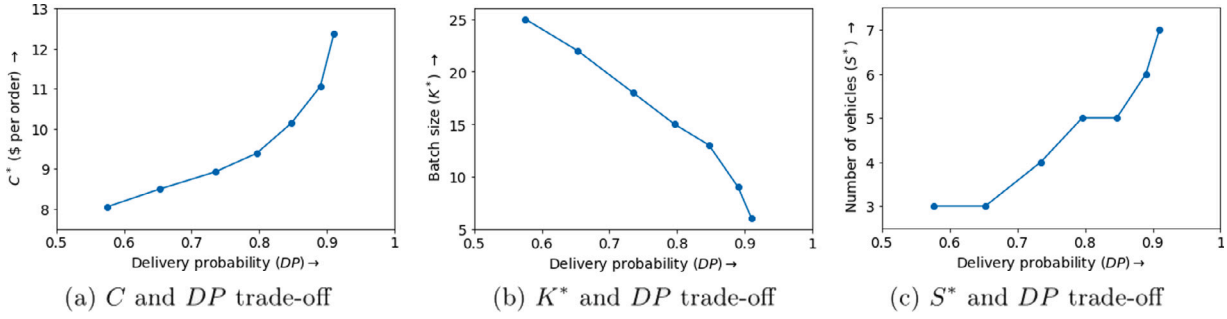


Fig. 10. Cost-quality trade-offs.

Table 6
Effect of warehouse layout, order batch size, and picker and vehicle utilization on variance of order fulfillment times (it is low when both U_p and U_d are ≈50% and high when both U_p and U_d are ≈95%; service region layout (l', l) is 11 km, 11 km).

Warehouse layout	K	σ_r (min)		σ_b (min)		σ_q (min)		E_{fl} (min)	
		Low	High	Low	High	Low	High	Low	High
14*13	5	0.00	1.78	1.92	1.63	0.00	13.53	44.58	58.23
	10	0.00	1.78	3.46	3.11	0.00	14.29	47.33	61.73
	15	0.00	1.78	4.95	4.57	0.00	14.43	54.42	68.96
26*7	5	0.00	1.55	1.90	1.61	0.00	12.37	45.88	58.35
	10	0.00	1.55	3.45	3.09	0.00	13.78	48.79	62.65
	15	0.00	1.55	4.93	4.55	0.00	13.17	55.33	68.59
8*23	5	0.00	1.95	1.91	1.66	0.00	15.61	47.73	63.45
	10	0.00	1.95	3.46	3.15	0.00	15.20	51.00	66.33
	15	0.00	1.95	4.95	4.61	0.00	16.67	58.39	75.18

σ_b. At high picker utilization, c_{pb} decreases because the order inter-departure time from the pick process is dominated by the CV of order pick times. This leads to a generalizable observation that isolated sub-models of the order fulfillment process will not lead to the optimal setting of the design parameters. The upstream pick process affects the performance of downstream batching and delivery processes. Hence, a decision support tool is necessary for planning worker capacity and setting appropriate due times for delivery.

Due time setting for order fulfillment

In this section, we demonstrate how to set the due time for the order fulfillment process. To set the order fulfillment due time, we need to determine the mean and variance of order fulfillment time. Assume the fulfillment center wants to deliver 90% of the customer orders before the due time. The steps to set the due time for order fulfillment time (for given DP, N, K, and S) are as follows:

1. Estimate the mean and variance of order picking time using Eqs. (8) and (9).
2. Estimate the mean and variance of order batching time using Eqs. (10) and (11).
3. Estimate the mean and variance of order delivery time using Eqs. (15) and (16).
4. Use the normal distribution to compute the due time for batch fulfillment. The expression for due time is $E_{ft} + z\sqrt{V_{ft}}$, where z is the standard normal variate (for a 10% upper tail probability, z is 1.28). We relax the assumption of a normal distribution in Appendix C.

Now, we estimate the due time using the numerical experimentation data. The parameters related to the storage area and service region are as defined in Section 7.1. Table 7 shows the cost-optimal due time with a specified $DP = 90\%$ and $\lambda = 25$ orders per hour for different values of c_a. The results show that the realized due time corresponding to a DP metric depends on the values of N, K, and S. We observe that the increase in order arrival variability, increases the waiting time in order picking and delivery queues, which is compensated by increasing the

Table 7
Due time setting (with $z = 1.28$ corresponding to $DP = 90\%$).

c_a	N^*	K^*	S^*	C ($/order)	E_{fi} (min)	$\sqrt{V_{fi}}$ (min)	Due time (min)
0.5	6	6	7	12.3	43.2	8.5	53.1
1.0	6	6	7	12.3	45.5	8.5	56.5
1.5	7	6	8	14.0	46.4	9.3	58.3
2.0	8	6	9	15.6	48.0	10.1	60.9
2.5	9	6	10	17.2	49.7	11.7	64.7
3.0	10	6	12	19.6	50.3	14.0	68.2

resource level (number of pickers and delivery vehicles). These results emphasize the importance of variability in order arrival and fulfillment processes in estimating the promised due time for a given DP level.

7.3. Managerial insights

In this section, we explain two key managerial insights that are derived from the numerical experimentation results.

Mean–variance trade-offs exists for order fulfillment time

Our analysis shows that imposing a delivery service quality (DP) constraint increases the expected delivery costs as more resources are needed. Apparently, achieving delivery reliability comes at a cost. However, imposing a higher delivery service quality may lead to a decrease in expected order fulfillment time. Furthermore, the variance of order fulfillment time decreases substantially in comparison to the case without service quality constraints. Moreover, without such a constraint, DP is significantly lower, which may affect customer satisfaction and decrease long-term sales. Our model provides a tool for the fulfillment manager to analyze the trade-off between the cost and mean delivery time and delivery reliability. In Appendix E, we analyze the delivery time to be promised by the retailer to break even and maximize profitability. Our analysis shows that a shorter promised time maximizes the fulfilled demand, but retailers need to promise a longer fulfillment time to maximize profit. Specifically, for very short fulfillment times (less than 25 min in our example), retailers may find it challenging to achieve a break-even.

Integrated analysis of order fulfillment system matters

The proposed analytical model helps managers decide on the cost-optimal number of order pickers, the delivery batch size, and the number of delivery vehicles to achieve the desired service quality. Our numerical results show that increasing the number of delivery vehicles results in higher expected order fulfillment cost, but reduces expected order fulfillment time and the optimal delivery batch size. Therefore, isolated performance analysis of individual order fulfillment processes may not yield a global optimal solution. Hence, integrated analysis of the order fulfillment system using a decision support system is required to offer both cost efficient and reliable order fulfillment.

8. Conclusions and directions for future research

This study investigates the cost-quality trade-off in multi-line order fulfillment, consisting of order picking, order batching, and last-mile delivery. Studying these processes with an integrated model is important as they interact and jointly determine delivery performance. Few studies have examined these processes or investigated the impact of the variability of these order fulfillment processes on performance. We explicitly consider this variability in the order fulfillment processes and present an analytical model that integrates order picking, order batching, and order delivery.

First, we develop an analytical model to estimate the mean and variance of the order fulfillment sub-process and then integrate them using an open queuing network model. The integrated model captures the interaction between the processes and provides a framework to determine the mean and variance of the total order fulfillment time. We then develop a cost optimization model that helps in deciding the number of pickers, the delivery batch size, and the required number of delivery vehicles, subject to delivery service constraints (DP).

We provide several managerial insights, which help understand the trade-offs between expected order fulfillment cost and DP. The model without DP constraint gives lower expected order fulfillment cost and time, but leads to very high variance of order fulfillment time, and low DP. Focusing only on low fulfillment cost and mean fulfillment time can lead to lower delivery reliability and customer loss in the long run. However, for the case investigated (Instacart data in the San Francisco area), lowering the variance in fulfillment time comes at a considerable cost. We provide insight in the cost and service quality trade-off that helps managers to decide on their order fulfillment processes.

We also provide a tool that helps the manager decide on the delivery due time for each order. In the underlying model, we assume random storage positions in the warehouse and picking by single order. Such situations are typical in online grocery retail with a high number of line items. In the Instacart dataset, an average order comprises ten line items. We assume a normal distribution for the order fulfillment time, but the model can be easily adapted to any probability distribution. We assume that unlimited space is available at the picking and batching station. As we focus on quick orders only, for which the fulfillment time is very short, the retailer generally has a sufficiently large buffer space, so that waiting time due to limited buffer space for orders in the warehouse is avoided. However, when staging space is tight, this may still be modeled using a finite capacity queuing model, as done e.g. by Dallery and Frein (1993)

While we discuss the model in a q-commerce context, our model can be applied to set due times and decide optimal resource parameter settings for express deliveries too. Future research can relax some of the assumptions we made in our study. Orders may be picked by batch in the order picking process. Our current models may still be valid if a sort-while-pick strategy is adopted, assuming order batching has taken place offline. However, in a pick-and-sort strategy, the batch has to be formed before picking and the picker retrieves all the line items of a batch of orders instead of retrieving one customer order in a trip. This would require an additional order sorting process. It would be interesting to determine the order batch size before the order picking process and its effect on other order fulfillment resources and to integrate a downstream sorting process in the model. In our study, we analyze the steady-state performance measures, where the delivery batch size is static. However, it may be interesting to devise a dynamic batching policy based on the number of orders present at different fulfillment processes. For example, if the number of orders waiting for delivery is large, then it may be beneficial to increase the delivery batch size. Similarly, if the batch size is two and there is only one order waiting at batching, and the order picking station is empty, then it may be better to reduce the batch size to one instead of waiting for an order to complete the batch. We also assume random customer locations in the service area and the customer is available during the delivery slot. We can relax these assumptions. In addition, a batch of orders for delivery is made without considering the delivery locations in the service region. This assumption may also be relaxed by considering the batching of orders per delivery zone in the service region. It will also be interesting to study the case when pickers pick a few orders and then use their vehicles to deliver these orders to the customer, so there will be no order delivery queue. This business model is adopted by several q-commerce companies, including Instacart. In our study, we allow the demand to vary over time slots within a day. However, we still assume stationary demand within each time slot. Hence, it is an interesting avenue to research heavy demand fluctuations, when stationarity assumptions are no longer valid (see Chan et al. (2017), Liao et al. (2012), and Stolletz (2022) for details on optimizing time-dependent queuing systems), and to see how these demand fluctuations affect the integrated processes.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ejor.2024.03.003.

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